

Improvement of saturation magnetic flux density in Fe–Si–B–Nb–Cu nanocomposite alloys by magnetic field annealing

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ABSTRACT

In the present study, we investigated the effect of heat treatment on the magnetic properties of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) alloy system. Ribbon samples with the width of 2 mm and thickness of 20–30 μm were synthesized by melt spinning. Among the samples, only amorphous samples were nanocrystallized by annealing between 360 and 600 $^{\circ}\text{C}$. The annealed alloys tended to have improved soft magnetic characteristics, including higher saturation magnetic flux density (B_s), while maintaining high relative permeability (μ_r), and low core loss (P_{cv}). A high saturation magnetic flux density of 1.51 T was obtained for the $\text{Fe}_{78}\text{Si}_{19}\text{B}_9\text{Nb}_3\text{Cu}_1$ core sample after longitudinal magnetic field annealing at 520 $^{\circ}\text{C}$. The sample had excellent soft magnetic properties as compared to the conventional value of FINEMET core ($B_s = 1.23$ T), which is a significant improvement as compared to the as-spun ribbons ($B_s = 1.10$ – 1.26 T).

1. Introduction

Over the past several decades, nanocrystalline materials have attracted a considerable scientific and technological attention due to remarkable soft magnetic properties of these materials. These materials are widely used in electric power applications, such as transformer cores and other inductive devices. In recent years, an important concern has been energy saving. Electrical power supplies require a low iron loss and a high saturation magnetic flux density (B_s) in soft magnetic materials. From this viewpoint, the Fe–Cu–Nb–Si–B nanocrystalline alloy ribbons with a low coercivity (H_c) and core loss (P_{cv}) have been developed as novel soft magnetic materials, which are neither Si steels nor Fe-based amorphous alloys.

From then on, three typical nanocrystalline alloy systems have been developed under the trade name of FINEMET (Hitachi Metals Ltd., Japan) (Yoshizawa et al., 1988), Nanoperm (MH&W International Corp., USA) (Suzuki et al., 1991a, 1991b) and Hitperm (Carnegie Mellon University, USA) (Willard et al., 1998). Among these alloys, Nanoperm exhibits high B_s of 1.5–1.7 T but contains expensive and easily oxidized elements (e.g. Zr or Hf). Hitperm possesses high Curie temperature but the H_c is up to 200 A/m and the large amount of additive element Co increases material cost [1]. Compared to the above alloys, FINEMET exhibits excellent soft magnetic properties, as well as convenient production process with economical cost. The nanocrystalline FINEMET™ alloy was first produced by Yoshizawa in 1988 [2–4].

This alloy has a particular microstructure characterized by randomly oriented, ultrafine α -FeSi grains with a bcc structure [5]. These characteristics constitute the basis for the outstanding soft magnetic properties of the alloy, with a high level of initial permeability ($\sim 10^5$) and a low saturation magnetostriction ($\sim 2 \times 10^{-6}$) with a relatively good B_s , of about 1.23 T [6]. The premier soft magnetic characteristics are based on the two-phase microstructure consisting of nanocrystalline ferromagnetic grains surrounded by a ferromagnetic amorphous matrix. In order to obtain optimal microstructure and properties, amorphous ribbon precursors produced by melt spinning are generally annealed at a certain temperature above the primary crystallization temperature (T_x). On the other hand, the relative permeability (μ_r) as well as P_{cv} and B_s depends on the presence of magnetic field in annealing. In particular, the annealing under longitudinal magnetic field results in squared hysteresis loops characterized by a significant reduction of H_c [7]. In actual applications such as saturable reactors or choke coils, the soft magnetic materials which show high remanence or flat B–H curves are preferably used. Therefore, the magnetic field annealing could be a useful method for improving the magnetic properties of amorphous alloys [8].

Therefore, in the present study, we successfully developed new Fe–Si–B–Nb–Cu nanocrystalline soft magnetic alloys by magnetic field annealing. In the present study, we obtained nanocomposite soft magnetic alloys with a high B_s , while maintaining their good magnetic properties with the replacement of Si by Fe.

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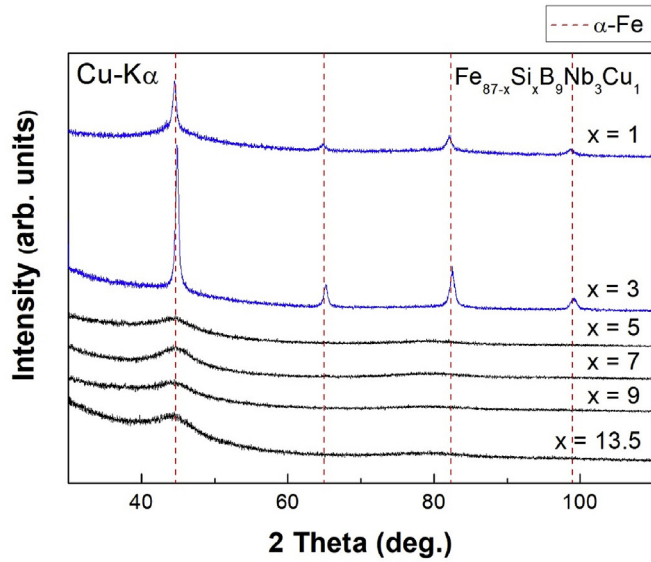


Fig. 1. The XRD patterns of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) as-spun ribbons.

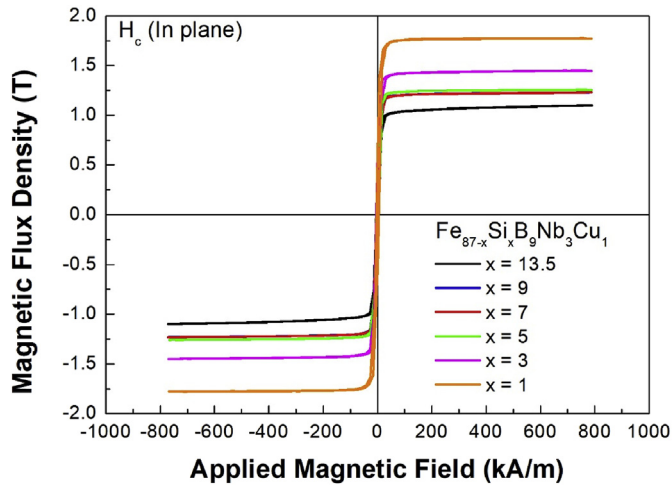


Fig. 2. Hysteresis loops of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) as-spun ribbons.

Table 1

Saturation magnetic flux densities, structures, and compositions of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) as-spun ribbons.

Composition (at.%)					Structure	B_s (T)	
Fe	Si	B	Nb	Cu			
73.5	13.5	9	3	1	Amorphous	1.10	FINEMET
78	9	9	3	1	Amorphous	1.23	
80	7	9	3	1	Amorphous	1.23	
82	5	9	3	1	Amorphous	1.26	
84	3	9	3	1	bcc-Fe	1.45	
86	1	9	3	1	bcc-Fe	1.77	

2. Experiments

Alloy ingots with a composition of $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) were prepared by arc-melting mixtures of high-purity Fe (99.95%), Si (99.999%), B (99.5%), Nb (99.95%), and Cu (99.99%) under the Ti-gettered argon atmosphere. All ingots were re-melted four times to properly mix the elements constituting the material.

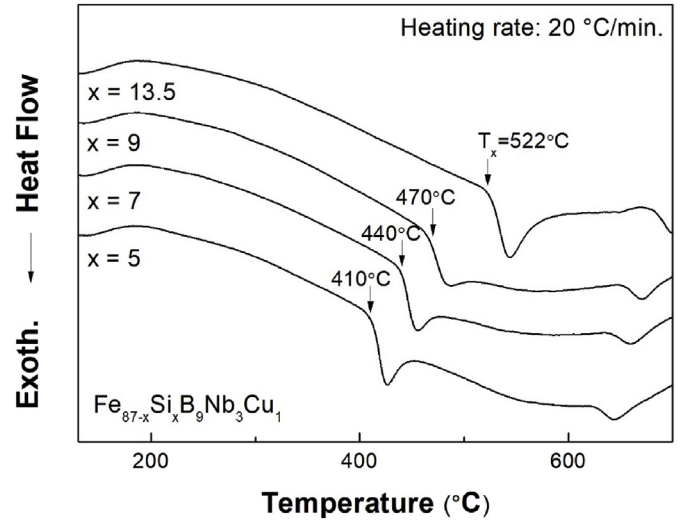


Fig. 3. The DSC curves of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 5, 7, 9$, and 13.5) as-spun ribbons.

Table 2

Annealing temperature, crystallization temperatures, and compositions of the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 5, 7, 9$, and 13.5) as-spun ribbons.

Composition (at.%)					T_x (°C)	Annealing temperature (°C), 1h			
Fe	Si	B	Nb	Cu					
73.5	13.5	9	3	1	522	450	500	550	600
78	9	9	3	1	470	420	470	520	570
80	7	9	3	1	440	390	440	490	540
82	5	9	3	1	410	360	410	460	510

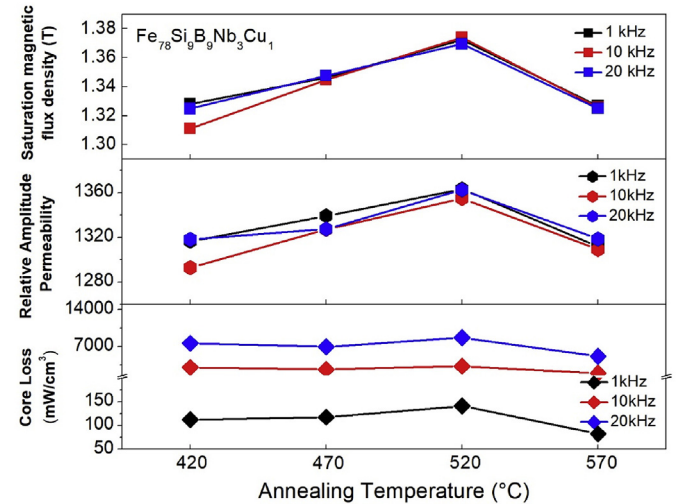


Fig. 4. Variation of B_s , μ_r , and P_{cv} values in the annealed ribbons of $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ at 1, 10, and 20 kHz for 60 min with increasing T_a .

Amorphous alloys were produced in the form of ribbons with the thickness of 20–30 μm and width of 2 mm by using the melt spinning under argon atmosphere at the roll speed of 39.3 m s^{-1} . The as-spun samples were subjected to annealing treatments at various temperatures above T_x for 60 min under the argon atmosphere. The structures of as-spun and annealed samples were examined by X-ray diffraction (XRD) with Cu-K α radiation and transmission electron microscopy (TEM). Thermal properties associated with the T_x were investigated by differential scanning calorimetry (DSC) under the flowing argon atmosphere and constant heating rate of 0.34 K s^{-1} . The hysteresis loops

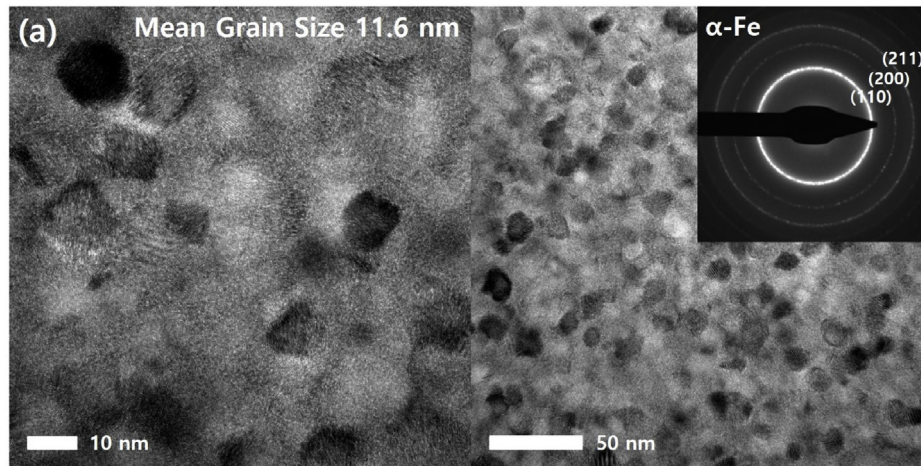


Fig. 5. Bright-field TEM image (left), selected area diffraction pattern (top-right) of the α -Fe phase, and the low magnification TEM image (right) of $\text{Fe}_{73.5}\text{Si}_{13.5}\text{B}_9\text{Nb}_3\text{Cu}_1$ ribbons annealed at (a) 500, (b) 550, and (c) 600 °C.

of as-spun and annealed samples were measured at room temperature with a vibrating sample magnetometer (VSM) under the in-plane applied magnetic field ranging from $-10,000$ to $10,000$ Oe. In addition, the magnetic properties of B_s , H_c , μ_r and P_{cv} were examined using an AC B–H loop tracer. The B_s , μ_r and H_c were measured under the maximum applied field (H_m) of 800 A/m, and the P_{cv} was measured under f of 100 kHz, and the H_m of 0.1 T.

3. Results and discussion

Fig. 1 shows the XRD patterns for the as-spun ribbons of $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5). The patterns consisted of a broad halo hump, indicating the formation of the amorphous phase. No diffraction sharp peak corresponding to a crystalline phase was observed for $x = 5, 7, 9$, and 13.5 . As can be seen in Fig. 1, when the Fe content exceeded 84 at.%, the as-quenched samples were heavily crystallized, and these sharp peaks corresponded to α -Fe nanocrystals.

Fig. 2 depicts the hysteresis loops for the $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) as-spun ribbons. All loops exhibited typical soft magnetic behaviors. The high saturation magnetization (M_s) of the amorphous ribbons was measured using VSM; the values of M_s were summarized in Table 1. As can be seen in Fig. 2, M_s increased from 1.10 T to 1.26 T with the decrease of Si content from 13.5 to 5 at.%. The other two alloys ($x = 1$ and 3) show high M_s , because both contain bcc phases. It is assumed that the volume fraction of the bcc phase that have higher magnetization than that of the amorphous phase decreases with

increase in Fe content for the ribbons with the mixed structure [9]. It is well known that the appropriate crystallization volume fraction leads to the compensation of magnetostriction of bcc-Fe phase and of the remaining amorphous one [10].

DSC under the flowing argon atmosphere is necessary to search the annealing temperature which should be higher than T_x . The DSC curves which indicate the crystallization behavior of as spun ribbons with $x = 5, 7, 9$, and 13.5 are shown in Fig. 3. The exothermic peaks observed in the two-stage crystallization process in the DSC curve are related to the crystallization sequence of the amorphous alloys. These peaks could be attributed either to the further crystallization of the residual amorphous phase, or to phase transformation of the existing metastable phases, such as bcc-Fe, occurring after the primary crystallization. Based on the DSC results, the annealing temperature values, T_a , were in the temperature range between the end of the first stage and the start of the second stage, so that only the α -Fe phase would precipitate during annealing.

In order to achieve the nanocrystalline state, the ribbons were submitted to annealing treatments under the argon atmosphere at various temperatures in the vicinity of T_x . The annealing temperatures are summarized in Table 2.

The annealing temperature dependence of P_{cv} , μ_r , and B_s for the amorphous and partially crystallized $x = 9$ sample are shown in Fig. 4. As can be seen in Fig. 4, B_s considerably increased with an increase of T_a . Furthermore, B_s reached the maximum value of 1.37 T, then decreased at $T_a = 570$ °C. The μ_r showed a variation tendency similar that

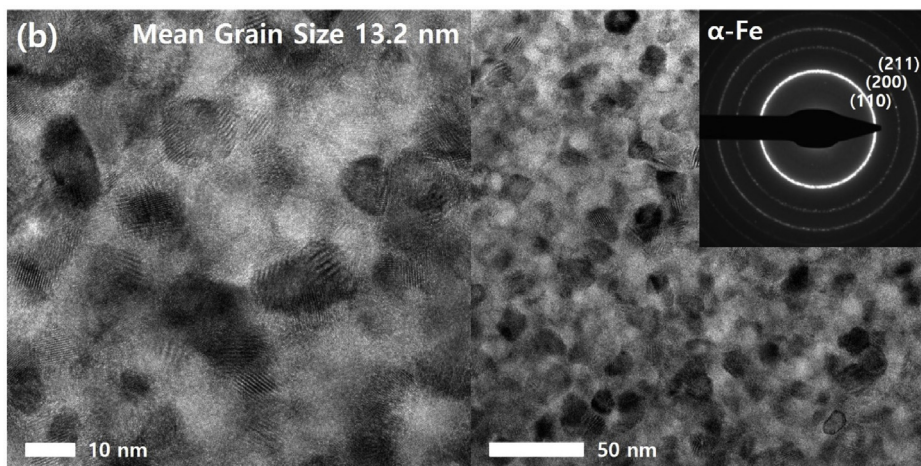


Fig. 5. (continued)

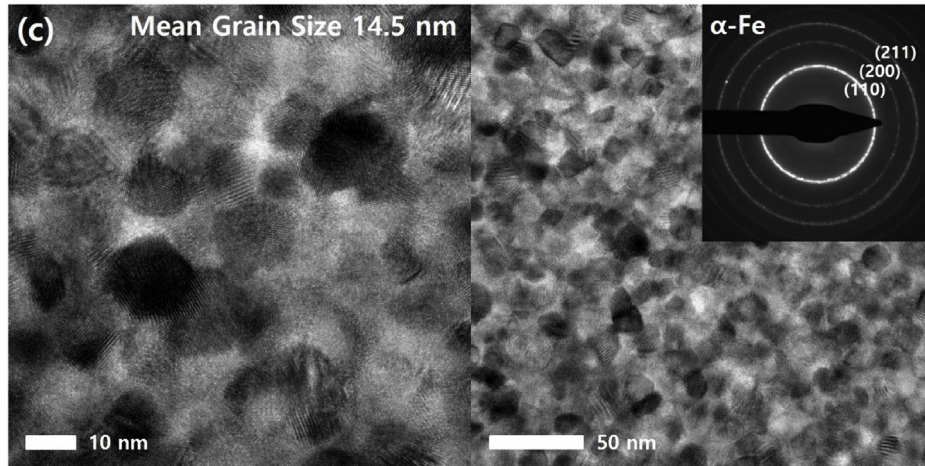


Fig. 5. (continued)

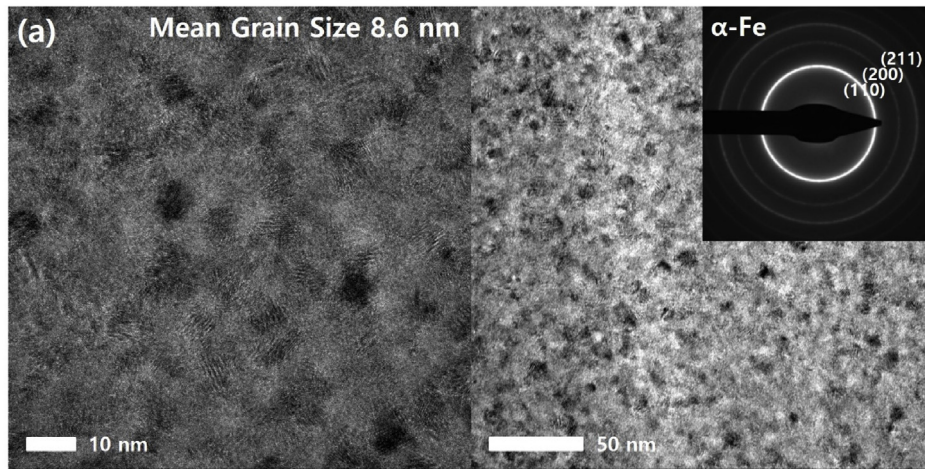


Fig. 6. Bright-field TEM image (left), selected area diffraction pattern (top-right) of the α -Fe phase, and the low magnification TEM image (right) of $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ribbons annealed at (a) 470, (b) 520, and (c) 570 °C.

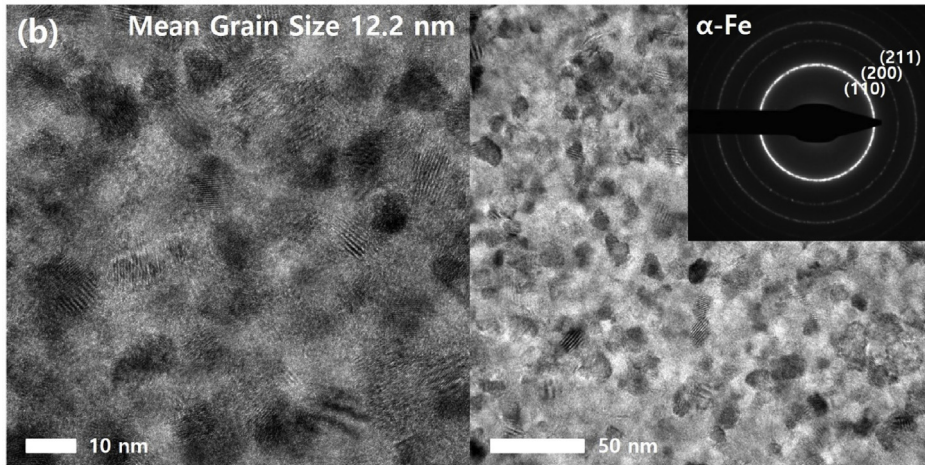


Fig. 6. (continued)

observed in B_s ; however, P_{cv} remained remarkably stable with the increase of T_a . The values of P_{cv} measured at 1 kHz retained a relatively low value as compared to those measured at 10 and 20 kHz; the minimum value of P_{cv} was 82 mW/cm^3 at $T_a = 570^\circ\text{C}$.

The changes in microstructure upon annealing were examined by TEM. Fig. 5(a)–(c) shows the TEM micrographs of differently heat

treated FINEMET samples ($x = 13.5$). The sample annealed for 1 h at 500°C is an example of thermally relaxed residual stress. The ribbon annealed at 550°C for 1 h corresponds to the alloy with a low fraction of nanocrystallines. The presence of a higher amount of crystalline phase is clearly visible for the ribbon annealed at 600°C . According to the SAED pattern, the structure of nanocrystals confirmed to be bcc. Typical

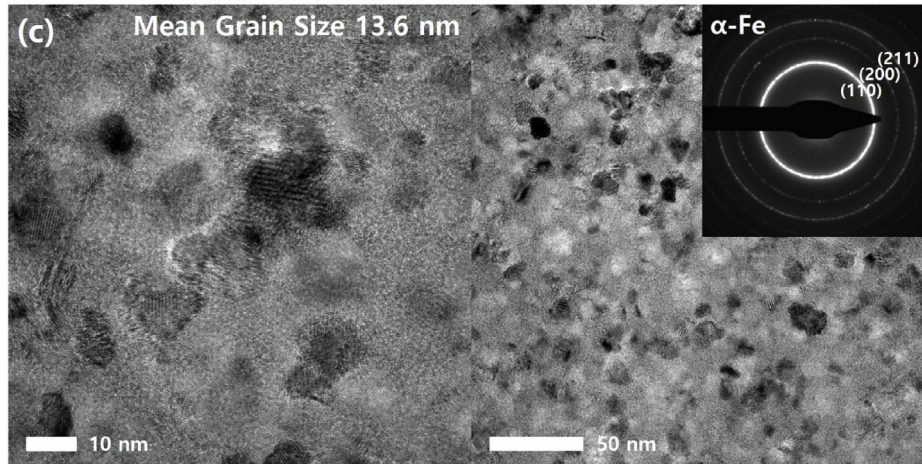
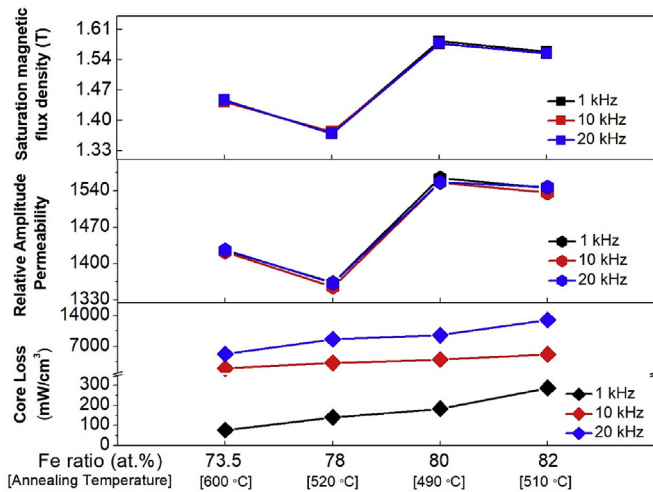


Fig. 6. (continued)

Fig. 7. Variation of B_s , μ_r , and P_{cv} values of the ribbons annealed at 1, 10, and 20 kHz for 60 min as Fe content of x at.%.

dimensions of ultrafine crystalline grains increased from 11.6 to 14.5 nm. This trend occurs because of the microstructure change from an amorphous to a crystalline structure caused by the precipitation of bcc-Fe phase and the changing volume fraction of the nanocrystalline phase. These increase in grain sizes resulted in improved of the value of μ_r and B_s , whereas the grain size is too large, the soft magnetic properties deteriorate due to the grain coarsening. Thus, the size of the Fe crystallites has been reported to be the most important feature [11–13].

As shown in Fig. 6(a)–(c), at the surface of the $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 9$) specimen, the Fe particles significantly grew by the increase in T_a from 470 to 570 °C. Typical dimensions of ultrafine crystallites increased from 8.6 to 13.6 nm. The distribution of the particles in the specimen annealed from 470 to 570 °C appeared uniform with no noticeable segregation, resulting in improvement of B_s of $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$.

Fig. 7 shows the values of P_{cv} , μ_r , and B_s with Fe content of x at.%. The graph shows an interesting trend that the tendencies of B_s and μ_r are similar. The $\text{Fe}_{80}\text{Si}_7\text{B}_9\text{Nb}_3\text{Cu}_1$ represented the excellent soft magnetic properties such as high B_s of 1.57 T. With an increasing Fe substitution for Si, the B_s and μ_r increased fairly linearly, but the $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 9$) deviated from that trend. Therefore, this temperature was selected to study the influence of magnetic field annealing on the soft magnetic properties. We additionally conducted heat treatment under magnetic field to improve the B_s and μ_r values.

To this end, the $\text{Fe}_{73.5}\text{Si}_{13.5}\text{B}_9\text{Nb}_3\text{Cu}_1$ (FINEMET for reference) and

$\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 9$) ribbon samples were coiled into a toroidal core with the outer diameter of 13 mm and the inner diameter of 8 mm using a core winding machine, and the cores were annealed in a longitudinal magnetic field. The cores were annealed under the longitudinal applied magnetic field of 0.15 T at 520 and 570 °C for 60 min to improve the soft magnetic properties. Properties of these annealed cores are characterized by two basic parameters: (1) Presence or absence of magnetic field and (2) T_a . The results of the cores annealed at 520 °C are shown in Fig. 8(a)–(c). The cores annealed under the presence of longitudinal magnetic field results in squarish hysteresis loops characterized by a significant reduction of the coercive field. The cores annealed in the longitudinal magnetic field exhibit an advanced degree of B_s and H_c as compared to the ribbons annealed in the non-magnetic field. The magnetic properties of $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 9$) cores, including annealing conditions, are summarized in Table 3. The H_c values of the specimens annealed at $T_a = 520$ °C significantly decreased from 57.728 to 11.270 at 1 kHz, from 121.32 to 18.294 at 10 kHz, and from 152.74 to 22.910 at 20 kHz. Typical B_s for the thermally relaxed by non-magnetic field annealing at 520 °C samples were around 1.37 T, while those for the samples annealed in the longitudinal magnetic field had the average value of 1.51 T below 20 kHz. These soft magnetic properties (such as high B_s of 1.51 T) are excellent as compared to the conventional value of FINEMET core ($B_s = 1.23$ T) [6].

4. Conclusion

In the present study, a controllable longitudinal magnetic field annealing was conducted in a series of amorphous and nanocrystalline $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) samples under various conditions. The effects of Fe substitution in the Fe–Si–B–Nb–Cu amorphous alloys on the soft magnetic properties were investigated from the viewpoint of nanocrystallization. The main results can be summarized as follows.

- (1) For $\text{Fe}_{87-x}\text{Si}_x\text{B}_9\text{Nb}_3\text{Cu}_1$ ($x = 1, 3, 5, 7, 9$, and 13.5) alloys, the amorphous phase remained almost unchanged with Fe content higher than 5 at.%, but clearly changed to a mostly single bcc phase at Fe contents lower than 3 at.%.
- (2) The bcc-Fe particle size significantly increased by the annealing temperature, T_a . These decrease of volume fraction result in fairly improve the value of M_s . For example, the $\text{Fe}_{80}\text{Si}_7\text{B}_9\text{Nb}_3\text{Cu}_1$ represented the excellent soft magnetic properties such as high B_s of 1.57 T.
- (3) The B_s , μ_r , H_c and P_{cv} values of the specimens annealed under longitudinal magnetic field considerably improved as compared to those of the non-magnetic field annealed specimens. In addition,

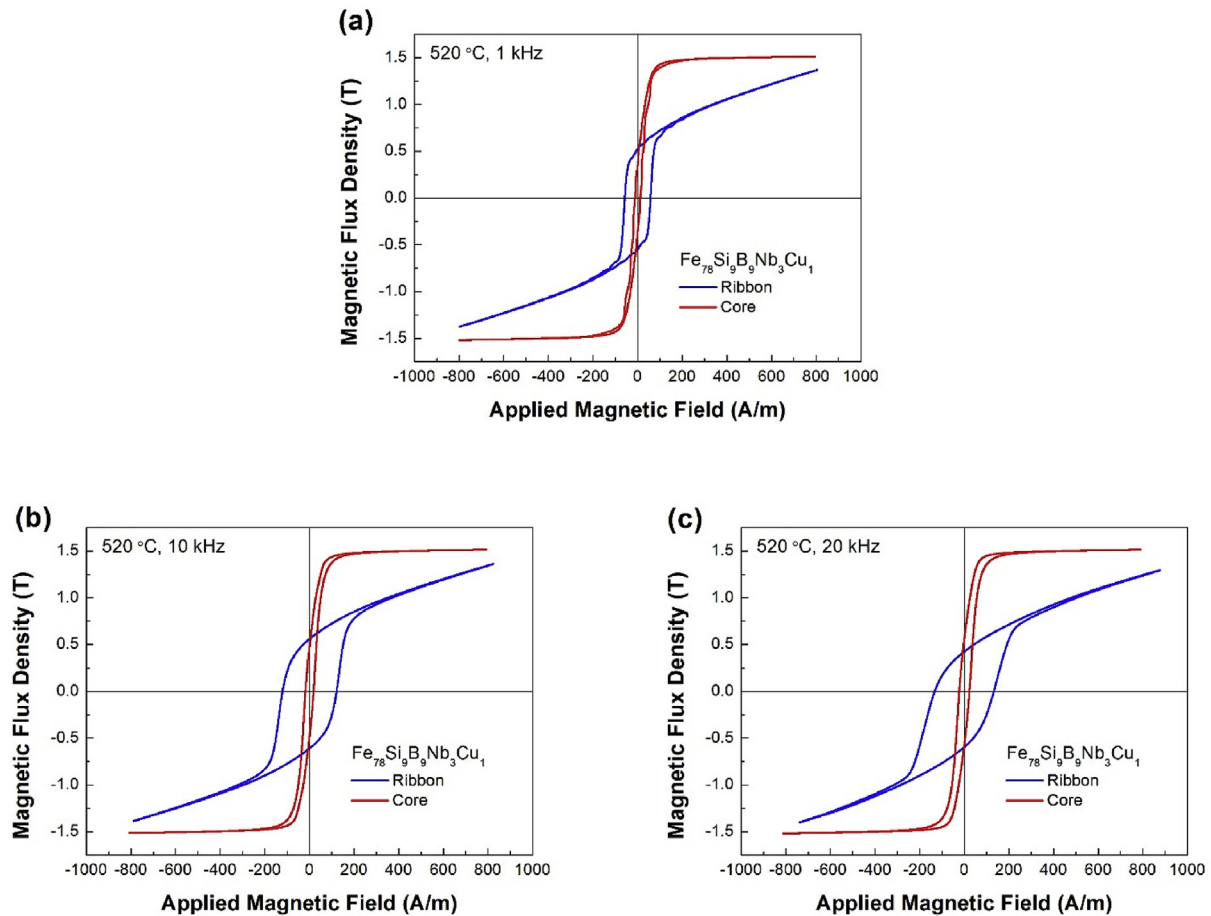


Fig. 8. Hysteresis B–H loops of $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ ($T_a = 520^\circ\text{C}$) ribbons and cores acquired via an AC B–H loop tracer at 1, 10, and 20 kHz.

Table 3

Summary of saturation magnetic flux density (B_s), coercivity (H_c), relative permeability (μ_r), and core loss (P_{cv}) for $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ cores at 1, 10, 20, and 100 kHz with $T_a = 520$ and 570°C .

$\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$		$T_a = 520^\circ\text{C}$		$T_a = 570^\circ\text{C}$	
		@ Non-magnetic field	@ Magnetic field	@ Non-magnetic field	@ Magnetic field
B_s (T)	1 kHz	1.3722	1.5147	1.3268	1.3350
	10 kHz	1.3739	1.5147	1.3257	1.3345
	20 kHz	1.3693	1.5165	1.3249	1.3376
	100 kHz	–	1.3993	–	1.1885
μ_r	1 kHz	1363.3	1511.5	1311.7	1331.8
	10 kHz	1354.9	1506.7	1309.0	1326.8
	20 kHz	1362.6	1506.2	1318.5	1327.3
	100 kHz	–	1387.2	–	1113.6
P_{cv} (mW/cm ³)	1 kHz	140.95	50.860	82.239	50.994
	10 kHz	3258.3	856.50	1936.3	937.38
	20 kHz	8639.2	2354.3	5140.4	2698.4
	100 kHz	–	31740	–	33830
H_c (A/m)	1 kHz	57.728	11.270	40.049	41.544
	10 kHz	121.32	18.294	78.465	47.407
	20 kHz	152.74	22.910	96.125	51.191
	100 kHz	–	56.000	–	72.151

the results suggest that a good compromise of B_s , H_c , μ_r , and P_{cv} values was obtained in the core annealed at $T_a = 520^\circ\text{C}$. The proposed alloy system, $\text{Fe}_{78}\text{Si}_9\text{B}_9\text{Nb}_3\text{Cu}_1$ produces excellent soft

magnetic properties, such as high B_s of 1.51 T as compared to the conventional value of FINEMET core ($B_s = 1.23$ T).

The new level of B_s obtained in this particular composition is considerably different from previous results and opens a novel path to new technologies for common mode choke or inductance applications.

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